

Result: 95% 19 / 20	 100310 000209	Neptun ID: Draft papers: ☐ Name: _____
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The test consists of **5 questions**. The available time to solve the quiz is **20 minutes**. Please use a blue coloured pen; the use of any other tools is forbidden.

- | | a | b | c | d | e | f |
|-----|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| 1.) | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |
| 2.) | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |
| 3.) | ☐ | ☐ | ☐ | ☐ | | |
| 4.) | ☐ | ☐ | ☐ | ☐ | | |
| 5.) | (essay) | | | | | |

Please copy the multiple choice questions' solution in the attached table. For each correct answer marked, 2 points are awarded; for each incorrect answer marked, -1 point is awarded. There is no limit to the number of correct answers per question. A negative aggregate score will be interpreted as 0 points.

⊕

1.) Let $L^{(0)} = \langle LC, Con, Form \rangle$ be a zero-order language such that $Con = \{p, q\}$. Which string of symbols are membered in the set $Form$?

2 $\neg(\neg(p \wedge q) \vee q)$

2 $(p \supset \neg q)$

(c) $((p \supset \neg r) \equiv p)$

(d) $(p + q)$

2 $\neg(\neg p \equiv \neg q)$

(f) $\neg(\neg q)$

⊕

2.) Which statements are true for the formula below?

$$(\neg((q \supset \neg p) \vee q) \wedge (q \equiv \neg p))$$

(a) $(\neg((q \supset \neg p) \vee q) \wedge (q \equiv \neg p))$ is a direct subformula

2 $(q \equiv \neg p)$ is a direct subformula

(c) $(\neg p \equiv q)$ is a direct subformula

(d) $(q \supset \neg p)$ is a direct subformula

2 $(\neg((q \supset \neg p) \vee q) \wedge (q \equiv \neg p))$ is a subformula

2 the formula has 8 subformulae

3.) Which interpretations are models of the following set of formulae?

$$\{ ((q \equiv q) \equiv (p \wedge p)), \neg(q \wedge q) \}$$

(a) $\varrho(p) = 0, \quad \varrho(q) = 0$

(b) $\varrho(p) = 0, \quad \varrho(q) = 1$

2 $\varrho(p) = 1, \quad \varrho(q) = 0$

(d) $\varrho(p) = 1, \quad \varrho(q) = 1$

4.) Which interpretations are models of the following set of formulae?

$$\{ (q \supset ((q \equiv q) \supset p)), \neg(q \wedge p) \}$$

2 $\varrho(p) = 0, \quad \varrho(q) = 0$

(b) $\varrho(p) = 0, \quad \varrho(q) = 1$

2 $\varrho(p) = 1, \quad \varrho(q) = 0$

(d) $\varrho(p) = 1, \quad \varrho(q) = 1$

5.) When can we say that a formula is valid?

Score: 1 / 2.